

Chukwudi Okereke Donald



Portfolio



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github.com/ChukwudiDonald

Education

University of Windsor

Bachelor of Computer Science

May 2022 - August 2026

Windsor, Ontario

- **Relevant Coursework:** Data Structures and Algorithms , Intro to Python, Intro to C & C++, Linear Algebra

Technical Skills

Languages: Python, C, C++, Java, HTML, CSS, JavaScript, SQL

Frameworks and Technologies: React.js, Django, Flask, FastAPI, Express.js, Node.js, Docker, Git, OpenCV, Pandas, NumPy, Matplotlib, MongoDB, FAISS, CUDA, Triton, MetaTrader5, MQL5, MQL4, OpenAI

Machine Learning: PyTorch, TensorFlow, Scikit-learn, LLMs, Transformers, RAG, LangChain

Experience

Research Assistant

Eminds Lab

Aug 2025 – Present

Windsor, Ontario

- Developed a Python library to automate serial data logging and real-time visualization, accelerating serial data collection and analysis process by 70%.
- Developed Flask APIs to concurrently collect and manage time-series data from multiple IoT devices with 100% accuracy.
- Developed firmware in C/C++ to collect micro-devices data and enable device communication via Wi-Fi, Bluetooth, and serial interfaces.

IT Support Clerk

Parklane Hotel

May 2024 – June 2024

Lagos, Nigeria

- Collaborated with the network administrator to test internet connections and resolve issues across hotel access points and routers ensuring reliable connectivity for guest and staffs .
- Coordinated with website and social media managers to align digital platforms with stakeholder requirements, successfully consolidating and removing redundancies across social media accounts.
- Directed the resolution of over 100 technical issues across electronic door and check-in systems, improving guest experience and operational stability.

Software Engineer Intern

Biodux

May 2021 – Aug 2021

Lagos, Nigeria

- Designed and implemented RESTful APIs for data processing and integration with frontend systems.
- Worked with relational databases to manage and query application data efficiently.
- Collaborated in an agile development environment, contributing to feature development and iterative improvements.

Projects

Hotel AI Chatbot | Python, REST API, RAG, LLM

- Developed a REST API that processes user queries and generates responses using a Retrieval-Augmented Generation (RAG) pipeline.
- Integrated external knowledge sources into LLM prompts to provide context-aware, accurate responses for hotel-related inquiries.
- Designed prompt injection logic to dynamically retrieve and incorporate relevant data, improving response relevance and consistency.

PySerialView | Python, Tkinter, Matplotlib, PySerial

- Developed a Python module for real-time visualization and logging of serial data from microcontrollers (e.g., ESP32, Arduino).
- Designed a responsive GUI using Tkinter with live plotting via Matplotlib for streaming sensor data.
- Implemented a producer-consumer architecture to handle continuous data streams without blocking the UI.